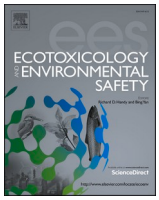
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Review



Comprehensive overview of triclosan neurotoxicity and construction of adverse outcome pathways using a systems toxicology approach: Triclosan-induced attention-deficit hyperactivity disorder as an example

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ABSTRACT

Triclosan (TCS) is widely applied to daily necessities as a chemical bacteriostatic agent. Environmental TCS levels have increased significantly following the coronavirus disease 2019 pandemic, posing a potential threat to humans and ecosystems. Epidemiological investigations and toxicological studies have shown that long-term TCS exposure can damage human tissues and organs and induce neurodevelopmental disorders in offspring. However, studies on the mechanisms underlying its neurotoxicity and toxicity risk assessments are limited. This review summarizes the status of environmental and human TCS exposure and systematically outlines its neurotoxic effects. To further elucidate the mechanisms underlying TCS neurotoxicity, using TCS-induced attention-deficit hyperactivity disorder (ADHD) as an example, we constructed an adverse outcome pathway (AOP) framework based on a systematic toxicology approach. We found that TCS increased the expression of cannabinoid receptor 1, which activates the "neuroactive ligand–receptor interaction" pathway, leading to ADHD-like behaviors, including cognitive, learning, and memory deficits, by modulating chemical synaptic transmission and neurotransmitter levels. The AOP framework was further used to assess the associated neurodevelopmental toxicity risks of TCS, contributing to a better understanding of its characteristics and safety. Thus, future research on the mechanisms underlying TCS toxicity and issues related to its detection and regulation should be emphasized.

1. Introduction

Triclosan (TCS; 5-chloro-2-[2′,4′-dichlorophenoxy]phenol), a synthetic broad-spectrum antimicrobial, was developed in the 1960s and is widely used in pharmaceuticals and personal care products (PPCPs); it was first licensed for use in 1964 by the Swiss company Ciba-Geigy and has rapidly become a ubiquitous substance to which humans are exposed (Windholz et al., 1978). As a Class III drug (a compound with high solubility and low permeability) classified by the Food and Drug Administration (FDA), TCS has been commonly added to toothpaste, facial cleansers, deodorants, soap bars, and anti-acne preparations and is clinically used in coated catheters, sutures, and dental adhesives to prevent postoperative wound infections (Neubert and Dillmann, 1972; Shrestha et al., 2020). In 2009, the American Public Health Association recommended discontinuing the use of TCS in PPCPs, and the United States Environmental Protection Agency (USEPA) listed TCS as a priority endocrine disruption in 2012. Considering the potential problems that may be associated with this type of antimicrobial agent, such as bacterial resistance, the FDA banned the addition of TCS to soap products (liquids, gels, foams, and bars) in September 2016; however, its use in toothpaste, mouthwash, and hand soap is still permitted (Dhillon et al., 2015). Canada launched a public comment period in December 2017 on the Pollution Prevention and Control Plan for TCS, which culminated in its inclusion in the Canadian Environmental Protection Act list of toxic substances, in July 2018. Although many countries have restricted the use of TCS, its consumption continues to increase. The worldwide

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production of TCS increased from 3 to 14 million pounds from 1998 to 2011 but decreased to 10.5 million pounds in 2015 (Weatherly and Gosse, 2017); however, its production rates have increased since the outbreak of coronavirus disease 2019 (COVID-19) owing to the high demand for disinfectants (Chu et al., 2020; Kumar et al., 2021).

TCS has been detected in the urine, blood, breast milk, and amniotic fluid of the general population worldwide (Etzel et al., 2017; Narrowe et al., 2015). Humans are exposed to this compound via dermal contact and direct ingestion (Narrowe et al., 2015; Yueh and Tukey, 2016). Personal care products containing TCS are recognized as its primary source. However, in addition to the direct use of TCS products, it is practically ubiquitous in the environment; exposure to TCS via air, surface water, drinking water, and soil is also possible and could lead to long-term exposure (Milanović et al., 2021). TCS has been classified as an environmental endocrine disruptor by the World Health Organization (Bergman et al., 2012), and growing evidence suggests that TCS has carcinogenic properties (Ashrap et al., 2017; Carey et al., 2015). As early as 2014, Mei-Fei Yueh (Yueh et al., 2014) first proposed that it may promote liver cancer. Further, the ovary is a potential target organ of TCS, and exposure to this compound can lead to apoptosis and oxidative stress in ovarian cells, which in turn promotes the occurrence of ovarian cancer (Kim et al., 2014). Additionally, many research suggests that TCS exposure result in various detrimental health effects. Specifically, TCS exposure has been demonstrated to induce histopathological alterations in the liver (Hemalatha et al., 2019; Regnault et al., 2018; Song et al., 2021), disrupt the diversity and composition of gut microbiota (Gaulke et al., 2016; Hirota et al., 2019; Narrowe et al., 2015; Restivo et al., 2021; Yang et al., 2018), and exhibit estrogenic activity (Huang et al., 2014; Ishibashi et al., 2004). Furthermore, TCS has been shown to interfere with thyroid function (Paul et al., 2012; Wu et al., 2016), elicit neurotoxic effects (Ruszkiewicz et al., 2017), and lead to obesity in offspring (Hua et al., 2018). Although TCS is widely present in the environment and increasing evidence has demonstrated its neurotoxicity, current studies remain fragmented and lack systematic integration. Therefore, a comprehensive review is urgently needed. It will help clarify the risks associated with environmental TCS exposure and its potential hazards to human health.

TCS is a neurotoxic agent (Iannetta et al., 2022; Ruszkiewicz et al., 2017) that can cross the blood–brain barrier (Van der Meer et al., 2017), affect the cognitive level of offspring (Kim et al., 2023), damage the hypothalamus and hippocampus, and cause morphological changes to the nervous system. Epidemiological studies have linked prenatal or perinatal TCS exposure to reduced attention, slower information processing, lower cognitive performance, and behavioral problems in children (Braun et al., 2017; Jackson-Browne et al., 2018; Cragoe et al., 2024). In animal models, maternal exposure to TCS causes neurodevelopmental disorders in the offspring and affects their neurobehavior (Diao et al., 2022; Wang et al., 2023). Our previous study further showed that TCS regulates ADHD-like behavior in male rat offspring by inducing dopamine deficiency in the prefrontal cortex (PFC). Moreover, increasing DA concentrations in the PFC can effectively prevent TCS-induced ADHD-like behavior (Cui et al., 2024). ADHD is one of the most common neurodevelopmental disorders (Wang, Liu, et al., 2017; Salari et al., 2023; Kooij et al., 2010; Polanczyk et al., 2007) and results from a combination of genetic and environmental factors (Regan et al., 2022). However, the effects and underlying mechanisms of environmental pollutant exposure on ADHD remain unclear, and the available evidence is still limited.

The Adverse Outcome Pathway (AOP) framework, a new tool for ecotoxicological studies, is a key component to measure human risk assessment of chemicals (Ankley et al., 2010; Vinken, 2013). The AOP framework consists of a molecular initiating event (MIE), a series of key events (KEs), and an adverse outcome (AO). Available evidence indicates that TCS is neurotoxic and may interfere with neurodevelopment in offspring. ADHD, one of the most common neurodevelopmental disorders in children, has attracted growing attention because of its potential association with environmental risk factors. Therefore, taking TCS-induced ADHD as an example, we employed the AOP framework to investigate the mechanisms underlying TCS-induced neurodevelopmental disorders in offspring. This framework integrates multi-level biological evidence and helps clarify the potential mechanisms of TCS-induced neurodevelopmental toxicity.

This study provides a comprehensive review of TCS contamination across different environmental media and its neurotoxicity. To investigate the potential mechanisms underlying TCS-induced neurodevelopmental toxicity, an AOP framework was constructed using ADHD as a representative outcome. Based on these findings, key challenges in the management of environmental TCS exposure are identified, and priorities for future research are highlighted.

2. Materials and methods

2.1. Literature search

We conducted a literature search using "Triclosan" as the primary keyword, identified relevant articles using the following keyword combinations: "environmental pollution," "water," "sediment," "soil," "air," "dust," "transformation," "degradation," "metabolism," "absorption," "exposure," "brain," "blood," "urine," "nervous system," "neurotoxicity," and "neurodevelopmental disorder." We combined these keywords using Boolean operators AND or OR when appropriate. The literature review covered studies from 2005 to 2025 published in PubMed and Web of Science.

A total of 374 articles were screened, of which 203 were included in the final analysis, whereas the remaining articles were excluded. The inclusion and exclusion criteria were defined as follows. The inclusion and exclusion criteria for the literature are presented as follows. Inclusion criteria: (1) Study type: original studies, including in vivo experiments, in vitro experiments, epidemiological investigations, and studies on toxicological mechanisms, with complete data and clear conclusions. (2) Language: only articles published in English or Chinese were included to minimize language bias. (3) Study subjects: studies involving TCS environmental exposure, neurotoxicity, neurodevelopmental disorders, or AOP construction in humans, mammals, or zebrafish, particularly those related to ADHD. (4) Content relevance: studies providing original data on TCS occurrence in environmental media, mechanisms of TCS-induced neurotoxicity, or chemical-induced neurodevelopmental disorders relevant to AOP construction.

Exclusion criteria: (1) non-original studies, such as conference abstracts, dissertations, book chapters, and review articles without original data; (2) studies irrelevant to the topic, including those focusing only on the antibacterial activity, synthesis, or other toxic effects of TCS without addressing neurotoxicity or environmental exposure; (3) studies with incomplete data, such as missing key information on TCS concentration, sample size, or experimental results, making data extraction impossible; and (4) duplicate publications reporting the same research content in multiple journals. During the literature screening process, two researchers independently developed the inclusion and exclusion criteria and excluded studies that clearly did not meet the requirements. Subsequently, the full texts of the preliminarily eligible articles were retrieved and reviewed, and the final selection was performed independently by the same two researchers.

2.2. AOP framework construction

The Comparative Toxicogenomics Database (CTD), GeneCards, DisGeNET, and MalaCards databases were selected for searches to collect genes and phenotypes associated with TCS and ADHD. Venn diagrams were used to identify the relevant genes, which were then subjected to Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway and Gene Ontology (GO) enrichment analyses on the Microbiotics website (https://www.bioinformatics.com.cn). The AOPs were constructed by the

OECD AOPs manual. The linkage between MIE and KEs was verified by previous literature to assess its applicability and credibility. Molecular docking was used to verify the combination of stressors and MIE.

2.3. AOP assessment

To verify the proposed KERs, relevant studies were retrieved from PubMed and Web of Science using combinations of the upstream and downstream event terms. The WoE was evaluated according to the OECD handbook based on biological plausibility, empirical support, and key event essentiality. KERs supported by multiple consistent studies and direct mechanistic evidence were rated as high, those supported by partial or indirect evidence were rated as moderate, and those with limited evidence were rated as low. Further details of the methods are provided in the Supplementary Information.

3. Results

3.1. Water

TCS is present in various water sources worldwide. This is a key concern in many surface water surveys. Its concentration is measured in oceans, rivers, lakes, wastewater treatment plants (WWTPs), and surrounding areas. The TCS concentrations in different water bodies, as well as in similar water bodies across various regions, were analyzed.

3.1.1. Surface water and groundwater

Rivers, lakes, and streams are important sources of drinking water. These water bodies are particularly susceptible to pollution, as they act as reservoirs for partially treated sewage and agricultural runoff, which contain high levels of TCS (Guruge et al., 2019). Fig. 1 shows the global distribution of TCS exposure in surface water and urine within various populations. The distribution map presents the basic exposure patterns, categorised by region and three population types.

The pollution in surface waters in Southeast Asia (India and Thailand), Africa (South Africa, Morocco, and Uganda), and South America (Brazil) is much higher than that in developed countries (Nozaki et al., 2023). River surveys conducted in the United States and Canada indicated that TCS concentrations are typically low, mostly below approximately 10 ng/L (Katz et al., 2013; Kumar et al., 2009; Lalonde et al., 2019; Wilson et al., 2009). European countries, such as Switzerland, the United Kingdom, Germany, Italy, and Spain, have the highest concentrations of TCS detected in surface water, below 100 ng/L (Bester, 2005; Gorga et al., 2015; Kasprzyk-Hordern et al., 2008; Lindström et al., 2002; Moldovan, 2006). Therefore, the exposure of surface water has been extensively studied (Table S1). The TCS concentration levels in rivers in various European countries are similar to those in Japan and South Korea and lower than those in some parts of the United States and most developing countries. Currently, research on TCS in surface waters in China has mainly focused on the middle and lower reaches of the Yangtze, Yellow, Pearl, and Hai River Basins. These levels were higher in the southeast and Pearl River watersheds and even higher in small rivers and residential areas. Algae plants in the water were the most sensitive to TCS, followed by invertebrates and fish. The Predicted No–Effect Concentration for combined aquatic species was 1.51 μg/L (Wang, Li, et al., 2024). Sampling results from WWTPs in Australia indicate that fish exhibit significant TCS exposure characteristics (Saaristo et al., 2024). TCS is a potent disruptor of fish hepatic

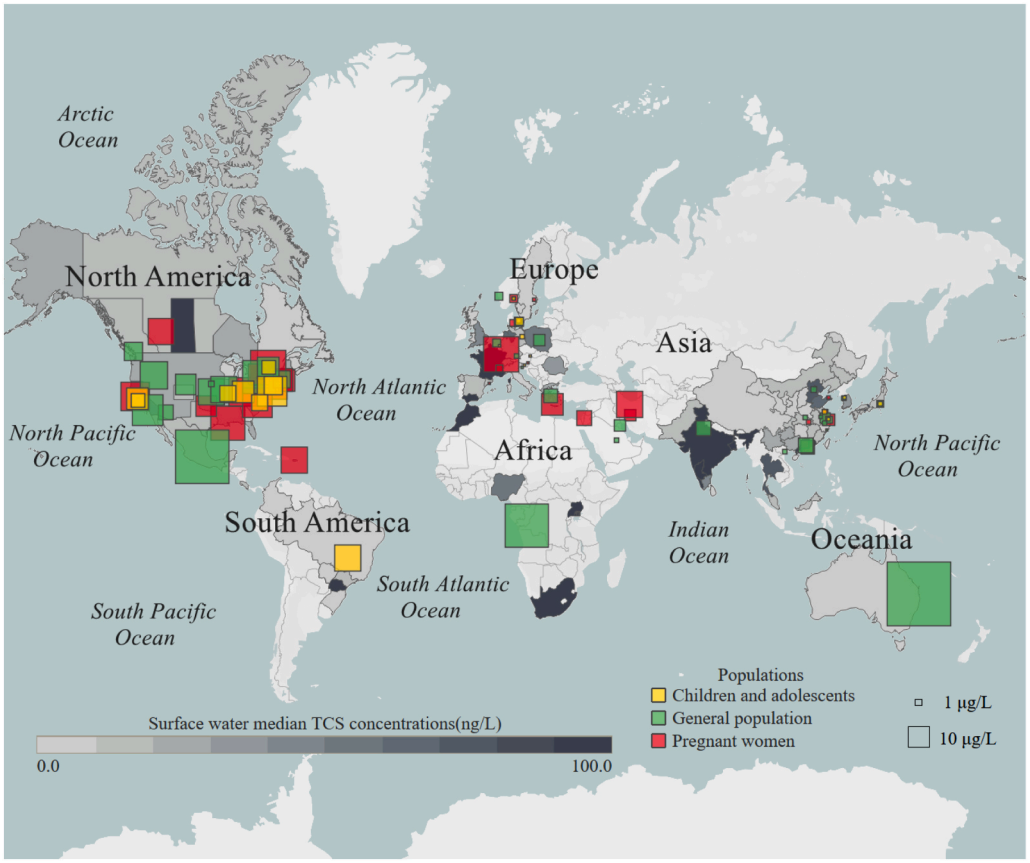


Fig. 1. Global profile of TCS levels among surface water and different populations. Different shades of gray indicate the concentration levels of TCS in surface water across different regions, with darker shading representing higher concentrations. Colored squares represent urinary TCS concentrations in different population groups, with larger squares indicating higher concentrations (yellow: children and adolescents; green: general population; red: pregnant women). The corresponding scales are shown in the figure. The data in Fig. 1 are from Tables S1 to S4.

development and metabolic homeostasis (Che et al., 2025). Furthermore, environmental TCS can disrupt lipid metabolism in the cyprinid fish *Squalidus argentatus* (Liu, Lu, et al., 2024). TCS severely disrupts the structure and function of subtropical freshwater ecosystems by inducing direct and indirect effects at multiple trophic levels, ultimately disrupting key biogeochemical processes (Yao et al., 2025). These findings indicate that TCS poses a potential ecological risk to local aquatic organisms and requires further attention.

The concentration of total TCS in surface water may be affected by various natural processes, such as known and unknown pollution sources, river flow (Wang et al., 2012), intensity of human activity (Zhao, Ying, et al., 2010), climate change (Goldsmith et al., 2020), and dilution, biotransformation, and degradation. The TCS contamination level in groundwater is generally lower than that in surface water (Milanović et al., 2021). This suggests that the aquifer system acts as a natural filter to remove and break down this compound from water (Vázquez-Tapia et al., 2022). Sources of groundwater pollution include surface infiltration, sewer leaks, and interactions with surface water (Ma et al., 2022). TCS in groundwater come from various sources, including urban and industrial wastewater, agricultural runoff, and waste disposal sites (Lapworth et al., 2012; Ma et al., 2022).

3.1.2. *WWTPs and drinking water*

Personal care products are the primary source of TCS in wastewater. Its concentration in untreated wastewater is high, resulting in a heavy environmental burden. A household survey in the city of Bathinda, India, estimated that the concentration of TCS in urban wastewater is one to two orders of magnitude higher than that in other countries and regions (Roy and Maurya, 2019). India conducted 2-year monitoring of the water quality of the Ganges River during the summer and winter seasons, and this study found that pollutant concentrations exhibit seasonal variations and pose significant ecological risks (Kantal et al., 2024). TCS concentrations in surface water samples in the United States, which are primarily composed of organic wastewater, were lower than those in influent samples from Canadian WWTPs, but significantly higher than those in influent samples from WWTPs in Switzerland and Germany (Guerra et al., 2019; Kolpin et al., 2002; Lindström et al., 2002; Wick et al., 2010). This situation may be related to the country's development status and the construction of urban wastewater systems.

WWPTs play a key role in maintaining the water quality of urban areas and rivers. Sampling points are set up upstream and downstream of WWPTs to detect TCS concentrations, which reflect their efficiency in removing TCS, and these are used to assess their wastewater treatment capacity. The TCS removal rate for American WWTPs is generally higher than 90% and can reach as high as 99% (Katz et al., 2013; Kumar et al., 2009). For example, the TCS concentration in Canadian WWTPs is reduced by at least one order of magnitude after wastewater treatment (Guerra et al., 2019). The TCS removal rates of five different types of WWTPs in Australia range from 72% to 93% (Ying and Kookana, 2007). The TCS removal rates in three WWTPs in Hong Kong, China, were improved through graded treatment processes, as shown by sampling analyses at each treatment stage (Wu et al., 2007). The TCS concentration detected in a WWTP in the Pearl River Basin in Guangzhou was lower than that obtained in a nearby WWTP in the Dong Jiang River Basin during the same year (Zhao, Ying, et al., 2010; Zhao et al., 2013). Changes in the TCS levels in wastewater are affected by the overall design, operating conditions, input load, treatment process, and climate. However, the TCS concentration in wastewater is significantly lower after multistage treatment than after primary treatment.

The wastewater treatment technologies used in different countries vary, and advanced treatment methods are relatively scarce in developing countries (Wee et al., 2020). In developed countries such as Australia (Ying and Kookana, 2007), South Korea (Yoon et al., 2010), Spain (Eulalia María Beltrán et al., 2020), Switzerland (Lindström et al., 2002), and Germany (Wick et al., 2010), TCS concentrations in effluent samples from WWTPs are generally low and exist in a narrow concentration range. In developing countries, such as Brazil, the downstream TCS concentration in WWTPs is higher, and the concentration range is wider (de de Almeida Brehm Goulart et al., 2021). This may be due to technical limitations resulting in low TCS removal rates, followed by the influence of other wastewater discharge.

Due to incomplete removal of TCS by WWTPs, TCS widely spread throughout surface water, groundwater, and drinking water through water reuse. The TCS concentrations detected in tap water in Malaysia (Wee et al., 2020), bottled water and tap water in Guangzhou (Li et al., 2010), tap water in Barcelona, Spain (Kantiani et al., 2008), and drinking water in various regions of the United States (Perez et al., 2013) are mostly below 10 ng/L. However, the highest concentration of TCS detected in drinking water in the United States is several tens of times higher than that in other regions (Perez et al., 2013). The detection rate of TCS in tap water is lower in Malaysia (30.50%) than in bottled water sold in China, Guangzhou (85.7%) (Li et al., 2010; Wee et al., 2020). Overall, the concentration of TCS detected in drinking water is relatively low, and the range is quite narrow.

Since the COVID-19 outbreak, the use of TCS-containing healthcare products has increased globally (Sun et al., 2024). As a widely used antibacterial disinfectant, the threat of TCS emissions to the environment has also become a focus of research (Xia et al., 2024). Kumar et al. first estimated TCS concentration variability in freshwater and marine environments and used toxicological data and biological models to assess their associated ecological risks (Kumar et al., 2024). TCS and its residues in the environment pose a threat to humans and other organisms (Jiang et al., 2024). The risk of human TCS exposure is primarily derived from tap and drinking water systems, and the implementation of removal technologies and toxicity reviews is imminent (Ebrahimi et al., 2023; Lee et al., 2024; Malihe Moazeni et al., 2023).

3.2. *Sediment and soil*

TCS concentrations have been detected in coastal sediments, WWTP effluent, activated sludge, and irrigation soils using activated sludge and reclaimed water (RW). The half-life of TCS in soil sediment under aerobic conditions is 21.52 days, and TCS in sediment alters oxidative stress and metabolic stress enzymes (Kantal et al., 2024).

The concentration and detection rate of TCS in coastal surface sediments are relatively low. Only 11% of coastal sediments in Zhejiang Province tested positive for TCS, with the highest concentration being 0.70 ng/g dry weight (Xie et al., 2022). In contrast, the coastal surface sediment detection rate for TCS is 44% in South Korea, with an average concentration of 0.41 ng/g dry weight (<0.1–6.10 ng/g dry weight) (Lee et al., 2019). These differences may be due to differences in human activities (e.g., river input, wastewater discharge, and aquaculture) in different coastal areas.

In soil environments, the primary source of TCS is activated sludge applied to the soil as a fertilizer following harmless treatment at WWTPs. Some water-scarce regions have also opted to use RW from WWTPs for crop irrigation. Beltrán et al. investigated the absorption and transportation of TCS by plants after irrigation with RW in urban areas of Spain. The residual concentration of TCS in the soil varies among different crops irrigated with RW (Eulalia María Beltrán et al., 2020). As crops absorb TCS from the soil, the compound enters the food chain, thereby increasing the risk of population exposure. Soil environmental studies have found that TCS has a higher tendency to adsorb onto microplastics (MPs) than onto soil, thereby increasing the risk of its migration in the soil. One of the sources of MP pollution in soil is the activated sludge obtained after wastewater treatment (Chen et al., 2021). TCS was detected in all activated sludge samples collected from 46 WWTPs across different regions of China, with a median concentration of 1680 ng/g dry weight (Zhu et al., 2019). In Canada, TCS was detected in 100% of 72 activated sludge samples collected from 13 WWTPs, with a median concentration of 8000 ng/g dry weight (Guerra et al., 2019). Moreover, the concentration of TCS in the activated sludge

samples from three WWTPs in Michigan ranged from 90 to 7060 ng/g (Cha and Cupples, 2009). Further, the TCS concentrations in the activated sludge samples from WWTPs in China are significantly lower than those in samples from WWTPs in the United States and Canada. Approximately 50% of activated sludge in the United States is used in land soils. Soil samples were collected from 10 different agricultural sites in 2007 and 2008 to determine the TCS concentrations. The concentrations of TCS were found to be 0.39 ng/g dry weight (0.16–1.02) and 0.16 ng/g dry weight (<0.05–0.28), respectively. Soil fertilized with activated sludge has much higher TCS concentrations. In the United States, the estimated TCS concentrations in soil improved with activated sludge are 340 ng/g dry weight for no-till soil and 4500 ng/g dry weight for tilled soil (Fuchsman et al., 2010; Verslycke et al., 2016). The application of activated sludge to soil increases the persistence of TCS and its accumulation in plant crops, resulting in an overall increase in the ecological toxicity risk (Fu et al., 2016).

The environmental concentration of TCS can influence the composition of microbial communities in sediments of different types and origins. The organic matter content in sediments may also modulate the effects of TCS on microbial communities (Wu et al., 2025). With the increasing applications of new materials, risks associated with the co-contamination of ecosystems with different chemicals cannot be overlooked. A recent study found that reduced graphene oxide (RGO) inhibits the accumulation of TCS in radish plants. Moreover, with RGO treatment, the key metabolite of TCS in soil—methyl TCS—decreases by 27.1%, indicating that RGO modifies the TCS degradation pathway, and its potential toxic risks warrant further exploration (Nie et al., 2025).

3.3. Air and dust

Considering the low volatility of TCS, its quality distribution in air is less than 1%, indicating that air is the least likely TCS migration pathway (Zhang et al., 2021). Currently, the number of studies on this topic is limited. Of note, the TCS detection rate in indoor dust samples collected across multiple studies has been reported as 100%, indicating that it is widely present in indoor air and dust, which are important sources of human exposure (Chen et al., 2018; Zhang et al., 2021; Zhu et al., 2020).

Indoor environments are often preferred for studying TCS concentrations in air. Chen et al. detected TCS in 80 dust samples collected from indoor air in American sports arenas and residences, with a median concentration of 390 ng/g (range, 20–3270 ng/g) (Chen et al., 2018). In contrast, residential dust samples from Belgium (median 220 ng/g, range 25–1830 ng/g), China (median 56.9 ng/g, range 3.83–3740 ng/g) and Vietnam (median 33.2 ng/g, range <5–1090 ng/g) contained much lower TCS concentrations (Geens et al., 2009; Tran et al., 2020; Zhu et al., 2020). The detection rate of indoor dust TCS in several cities of China are all above 95% , such as Shanghai (median 260 ng/g, range ND–1626 ng/g), Beijing (median 120 ng/g, range 40.1–1180 ng/g), Suizhou (median 24.5 ng/g, range <LOQ–730 ng/g) and Guangdong (median 20.7 ng/g, range <LOQ–127 ng/g) (Ao et al., 2017; Wang, Li, et al., 2021; Zhang et al., 2021). Meanwhile, a study in France found differences in TCS concentrations at different observation sites (residences, apartments, nurseries, and offices) (Laborie et al., 2016).

In this study, we found that the concentration of TCS in indoor dust samples in the United States is substantially higher than that reported for other countries and regions, indicating that concentrations in indoor air may be related, to some extent, to the degree of urban development. Higher concentrations of TCS were found in indoor-air in public places, such as sports grounds and offices, than in residential areas, possibly because of the frequent use of personal care products for disinfection and sterilization (Chen et al., 2018). In the living environment, TCS is mainly distributed in kitchen and bedroom dust, which may be related to the use of kitchen cleaning agents and personal care products (Tran et al., 2020). In this study, we paired participant urine samples with indoor dust samples and found a positive correlation between urine TCS levels and TCS concentrations in indoor dust (Wang, Li, et al., 2021). However, the effect of indoor dust intake on total TCS exposure is small (0.5%) (Wang, Li, et al., 2021; Zhang et al., 2021). Overall, these studies suggest that indoor dust exposure is an important route of human TCS exposure.

Timothy Abbott et al. have conducted extensive research on the persistence and bioaccumulation of TCS in the environment. Their research focused on the transformation and degradation products of TCS, as well as its fate in wastewater treatment and the environment (Abbott et al., 2020). Based on their findings, Fig. 2 illustrates the distribution and transformation of TCS and its metabolites in various environmental media.

3.4. Human exposure to TCS

Exposure to TCS occurs predominantly through consumer products that contain this chemical compound. Most TCS usage (up to 80%) is attributed to cosmetics, personal care products, and household cleaning agents (Gao et al., 2018). A recent study has suggested that human exposure to antibacterial food packets containing TCS is a potential health risk and may be a significant source of exposure to this compound (Klaiber et al., 2024). In addition to the direct use of TCS-containing products, exposure to TCS may also occur via air, surface water, drinking water, and soil, which could result in prolonged exposure. As depicted in Fig. 3, when humans ingest TCS, it is mainly absorbed in the oral mucosa and gastrointestinal tract (GIT). In humans, the concentration of TCS in the bloodstream peaks at 1–3 h after oral exposure. Moreover, once TCS enters systemic circulation, it can be detected in urine, plasma, and other biological samples (Sandborgh-Englund et al., 2006). The absorption rate of TCS through the skin is less than 10% (Queckenberg et al., 2009). A report shows that dermal administration of TCS increased the incidence of non-neoplastic epidermal lesions at the site of application (NTP Technical Report, 2024). TCS can be detected in various human biological samples, such as blood, breast milk, amniotic fluid, umbilical cord blood from newborns, adipose tissue, liver tissue, and brain tissue (Yang et al., 2021). A study in China showed that the detection rate of TCS in the serum of pregnant women can be as high as 96.3%, with the highest concentration of 5.84 ng/mL (Li et al., 2020). TCS can enter the human body through the skin and digestive tract, causing inflammation, disrupting thyroxine homeostasis, and interfering with glucose and lipid metabolism (Liu, Pu, et al., 2024; Ma et al., 2019). Long-term TCS exposure affects the growth of mice and induces substantive liver lesions, such as hepatitis and liver fibrosis (Liu et al., 2023; Peng et al., 2024). Chronic exposure also causes kidney damage and alters the intestinal microflora structure in adult mice (Liu, Lin, et al., 2024; Zhuang et al., 2024). In addition, TCS may increase the risk of hypertension during pregnancy and affect maternal and fetal health (Varshavsky et al., 2024). In the early stages, exposure can lead to the development of embryos with impaired mitochondrial function, altered steroid production, and disrupted metabolic activity (Papachristofi et al., 2025). TCS can also cross the placental and blood–brain barriers, posing a potential threat to early childhood development (Guo et al., 2020). Long-term exposure to low doses of TCS induces early pubertal onset, leading to precocious puberty in female mice (Gan et al., 2023). In addition to exposure to TCS alone, toxic effects induced by co-exposure to TCS and other substances have also been reported. The combined toxicity of TCS and polystyrene microplastics (PS) has been investigated in mice. Compared with single exposure, co-exposure to TCS and PS produced greater toxicity in the intestine and liver, which was characterized by intestinal barrier disruption, enhanced inflammatory responses and oxidative stress, hepatocellular vacuolar degeneration, and significantly elevated liver function indices. These findings support the risk amplification effect of co-exposure to two environmental pollutants (Liu et al., 2025). Zebrafish were also used as an experimental model to investigate the neurotoxicity of co-exposure to TCS and polypropylene (PP), polyethylene (PE), or polyvinyl chloride (PVC) microplastics.

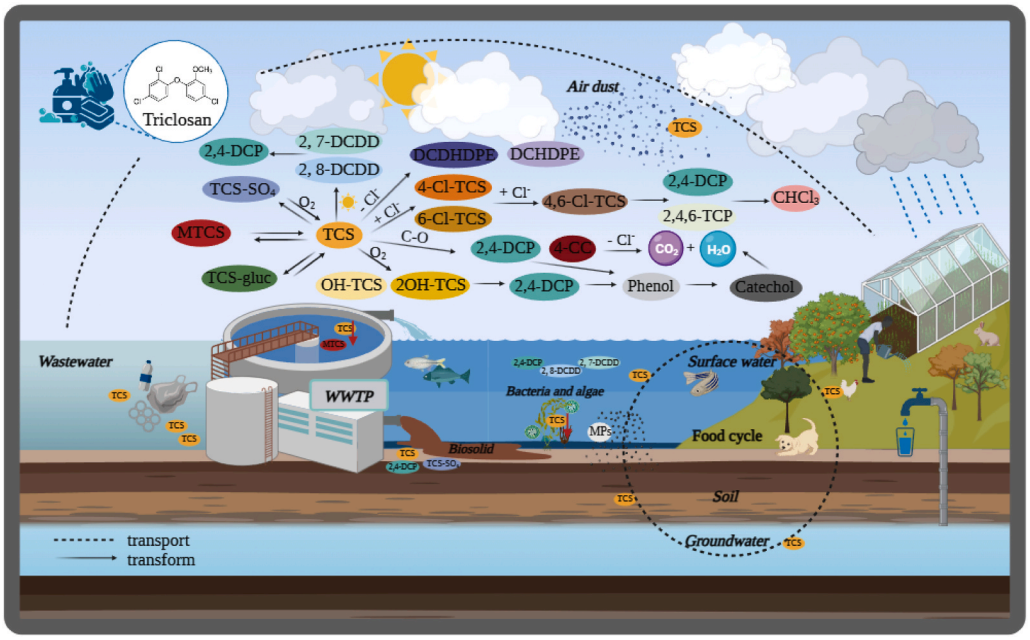


Fig. 2. Distribution and transformation of TCS and its metabolites in various environmental media. 2,4-DCP: 2,4-dichlorophenol; 2,4,6-TCP: 2,4,6-trichlorophenol; 2, 7-DCDD: 2, 7-dichlorodibenzo-p-dioxin; 2, 8-DCDD: 2, 8-dichlorodibenzo-p-dioxin; 4-CC: 4-Chlorocatechol; 4-Cl-TCS: 4,5-dichloro-2-(2,4-diclorophenoxy) phenol; 4,6-Cl-TCS: 4,5,6-trichloro-2-(2,4-diclorophenoxy)phenol; 6-Cl-TCS: 5,6-dichloro-2-(2,4-diclorophenoxy)phenol; 2OH-TCS: Dihydroxy-Triclosan; CHCl3: Trichloromethane; CT: Catechol; DCHDPE/DCDHDPE: Chlorinated hydroxylated diphenyl ether; H2QCl: 2-chlorohydroquinone; HQ: Hydroquinone; MTCS: Methyl Triclosan; MPs: Microplastics; OH-TCS: Monohydroxy-Triclosan; Ph: Phenol; TCS: Triclosan; TCS-gluc: Triclosan glucuronide; TCS-SO4: Triclosan sulfate; WWTP: Wastewater treatment plant.

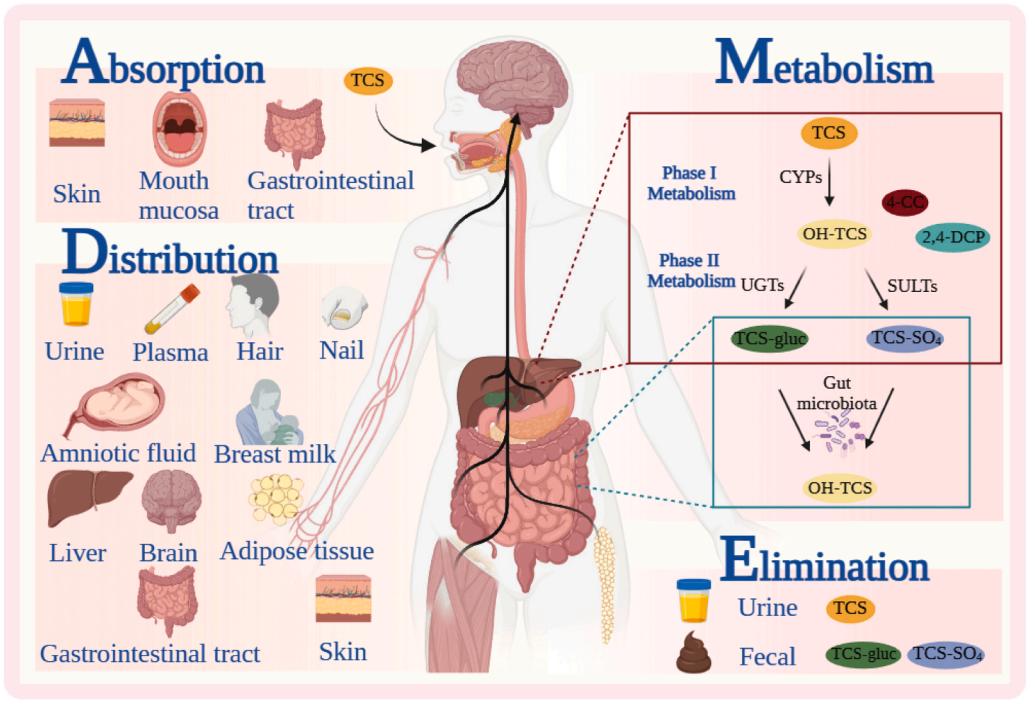


Fig. 3. The absorption, distribution, metabolism, and elimination properties of TCS in the human body. CYPs: Cytochrome P450; UGTs: UDP-glucuronosyltransferases; SULTs: Sulfotransferase.

Compared with TCS exposure alone, the combined exposure groups showed significant alterations in SOD activity, AChE activity, and 5-HT levels in zebrafish brain tissue. Among these groups, the TCS + PP group showed the strongest toxicity and aggravated neural injury by affecting amino acid metabolism, neuroactive ligand–receptor interaction, and other pathways. These results suggest that the combined toxicity was mainly synergistic (An and Zhang, 2023).

Human urinary specimens are widely available and commonly used for testing. The levels of urinary TCS in adults are greatly lower than the reference doses set by the USEPA and European Commission Scientific Committee on Consumer Products, which are 6400 and 2600 μg/L, respectively. Elevated urine concentrations of TCS have been observed

in children in southern China, whereas women, low-weight people, and youths are at a greater risk of exposure (Pan et al., 2024). In the United States, TCS is commonly detected in the urine of infants aged 6–12 weeks (Goldberg et al., 2024). The average concentration of TCS detected in urine samples of children was found to be 11 ng/mL (Jackson-Browne et al., 2020). Although TCS exposure varies among populations across different regions, its detection rates in urine samples are generally greater than 95%, indicating a high prevalence of human exposure to TCS (Liu et al., 2021; Wang, Li, et al., 2021; Yang et al., 2021). Table 1 provides the median urinary TCS concentrations for the general population, children and adolescents, and pregnant women in different countries and regions, categorised into four concentration levels.

According to extensive studies across three groups, TCS exposure in Asian and European countries is lower than that in North America. Although fewer studies have been conducted in South America, Africa, and Oceania, reported exposure levels are often higher. TCS exposure is substantially higher in developed countries (such as the United States and Canada) than in developing countries (such as China, India, and Iran). Moreover, TCS exposure was found to be higher in urban areas, such as Shenzhen and Shanghai, than in urban and rural areas of northern China, suggesting that exposure may remain higher in economically developed regions. This may reflect differences in product use intensity, population density, and related factors. However, this regional pattern is not consistent. Higher TCS exposure levels have also been reported in developing countries such as Brazil and the Democratic Republic of the Congo, possibly owing to technical limitations in WWTPs, as discussed above. These findings suggest that regional differences in TCS exposure cannot be explained solely by the level of economic development. In some low- and middle-income regions, elevated TCS concentrations may be associated with inadequate sewage collection systems and low wastewater treatment efficiency. These differences may have important implications for regional risk profiles. In addition, if the use of TCS in personal care products is not adequately regulated, direct human exposure may remain high. This could increase ecological and public health risks, even in regions with lower overall levels of industrial development than developed countries.

Table 1
Global profile of median urinary TCS levels among the general population, children and adolescents, and pregnant women.

Concentrations (μg/L)	General population	Pregnant women	Children and adolescents
<LOD	**Asia**: China rural regions (Zhejiang, Jiangsu). **Europe**: Slovenia.	**Europe**: Slovenia.	**Europe**: Slovenia.
<LOD-1	**Asia**: Northern China, Japan, Korea, Vietnam, Saudi Arabia. **Europe**: Germany.	**Asia**: China (Shandong, Jiangsu, Hubei). **Europe**: Sweden, Denmark.	**Asia**: China, Japan, Korea. **Europe**: Germany, Norway, Denmark, Belgium, Austria.
1–10	**Asia**: China (Guangzhou, Shenzhen, Shanghai), India. **Europe**: Greece, Norway, Kuwait, Poland, Belgium, Denmark. **North America**: Canada.	**Asia**: China (Guangdong, Shenzhen, Shanghai), Israel, Iran (Isfahan). **Europe**: Southeastern France, Norway.	**North America**: The United States, Canada.
> 10	**North America**: Mexico, The United States. **Oceania**: Australia. **Africa**: Democratic Republic of Congo.	**North America**: The United States, Canada. **Asia**: Iran (Tehran). **Europe**: Southwestern France, Greece.	**North America**: The United States (New Jersey). **South America**: Brazil.

Extensive research has been conducted on the exposure of pregnant women to TCS during different stages of pregnancy, with controversial findings. The Wuhan Women and Children Medical Care Center and Health Outcomes and Measures of the Environment detected a decrease in urinary TCS concentrations in pregnant women in late pregnancy; however, urine samples from pregnant women in the Puerto Rico Test site for Exploring Contamination Threats revealed higher detection rates and concentrations in the first and middle trimesters (Aker et al., 2019; Jackson-Browne et al., 2020; Jiang et al., 2019). Therefore, the relevance of the gestational period to TCS exposure outcomes in pregnant women is influenced by several factors.

Similarly, urinary TCS concentrations in Chinese children were found to be comparable, but the detection rate varied significantly, with a higher detection rate in 3-year-old children (99.5%) than in 7–11-year-old children (57.1%). A dynamic cohort study involving children in Shanghai, China, revealed that the younger the child, the higher the rate of TCS detection, indicating a negative association between the age of the child and the rate of TCS detection (Guo et al., 2020; Han et al., 2021). However, the results of a US study involving children aged 1–5 and 8 years showed an overall increase in TCS detection rates with an increasing age (Jackson-Browne et al., 2020), which is contrary to the findings of a Chinese study. Furthermore, a study conducted in Kerman, Iran, indicated that younger girls had higher urinary TCS concentrations than boys (Nasab et al., 2022), suggesting a correlation among child sex, age, and TCS exposure levels.

Besides, Several population-based reports have explored potential influence factors of exposure status, including age, gender, race-ethnicity, socioeconomic status, lifestyle, sampling patterns and timing, and consumption of PPCPs and cleaning products. However, in the context of the COVID-19 pandemic, which has increased the risk of TCS exposure, the long-term effects of TCS on neurodevelopmental outcomes remain unclear. Future research should focus on longitudinal cohort studies to provide stronger evidence for the adverse effects of TCS exposure on offspring cognition and behavior.

3.5. Temporal changes in TCS concentrations

To clarify the impact of increased disinfectant use during and after the COVID-19 pandemic on TCS concentrations in environmental and biological samples, this section summarizes the temporal changes in TCS levels in representative environmental media and biological matrices by comparing monitoring data collected before and after 2019. It further examines the impact of the pandemic on environmental and human exposure to TCS.

3.5.1. Environmental media

Before the COVID-19 pandemic, TCS concentrations in surface water worldwide were generally below 100 ng/L. In developed countries, such as the United States and Switzerland, median TCS concentrations in surface water were typically 1–10 ng/L, whereas in some developing regions, such as India and Brazil, they ranged from 10 to 50 ng/L (Katz et al., 2013; Guerra et al., 2019). After 2019, TCS concentrations in various environmental media showed a marked upward trend worldwide. For example, TCS concentrations in surface water in China ranged from 0.06 to 612 ng/L, with levels in the southeastern river basins and the Pearl River Basin generally falling within the range of 80–150 ng/L (Wang, Li, et al., 2024). TCS concentrations in the surface water of freshwater lakes have been reported to reach as high as 466 ng/L (Wang et al., 2022). Xia et al. (2024) investigated the occurrence of 25 emerging micropollutants associated with the COVID-19 pandemic in wastewater and reported that their concentrations generally increased in the post-pandemic period. Meanwhile, even in developed countries with high wastewater treatment efficiency, the predicted toxicity risk of TCS in urban surface waters increased significantly (Sun et al., 2024).

In sediments and soils, the median TCS concentration in global coastal sediments before the COVID-19 pandemic was generally below 0.5 ng/g dry weight (Lee et al., 2019; Xie et al., 2022). In Taihu Lake, sediment TCS concentrations ranged from 0.0058 to 8.3 ng/g, whereas in Baiyang Lake they ranged from 5.07 to 83.79 ng/g. In addition, post-2020 sampling of freshwater lakes in Wuhan, China, reported maximum sediment concentrations of 71 ng/g dry weight and 25 ng/g dry weight (Wang et al., 2022).

In air and dust, median TCS concentrations in indoor dust before the COVID-19 pandemic ranged from 20 to 390 ng/g, with the highest reported median concentration of 390 ng/g in the United States (Chen et al., 2018; Zhu et al., 2020). After 2019, the median concentration of $\sum$TCS + TCC in indoor dust from different cities in China was approximately 300–500 ng/g (Wang, Li, et al., 2021), which may be related to the frequent use of TCS-containing disinfectants during the pandemic. Overall, the currently available evidence indicates that TCS remains widely distributed across various environmental media, and that the COVID-19 pandemic may have altered its environmental inputs in certain regions. However, comparative data on TCS concentrations in environmental media before and after the pandemic remain limited. Therefore, further research is needed to clarify the spatiotemporal distribution of environmental TCS exposure levels.

3.5.2. Biological samples

Before the COVID-19 pandemic, the median urinary TCS concentration in the general population worldwide was generally below 10 μg/L. A study published after 2019 reported that TCS was detected in 99.5% of urine samples, with concentrations ranging from below the limit of detection (LOD) to 270 μg/L and a median concentration of 3.67 μg/L (Yang et al., 2021). Urinary TCS concentrations were higher in children than in adults (Pan et al., 2024). These findings suggest that human exposure to TCS may have increased after the COVID-19 pandemic. Meanwhile, a cohort study found that exposure to TCS may increase the risk of hypertension during pregnancy, suggesting that increased TCS exposure after the pandemic may pose a threat to maternal and infant health (Varshavsky et al., 2024).

Among 99 pregnant women in Iran after 28 weeks of gestation, TCS was detected in 100% of urine samples, with a median concentration of 2.89 μg/L, showing no significant difference from the pre-pandemic level (Attarian et al., 2023). A recent study reported that the detection rate of TCS in urine samples from pregnant women aged 18–42 years was 60.2%, with a median concentration of 0.49 ng/mL, suggesting a decrease in TCS exposure levels compared with previous years (Huang et al., 2025). However, it should also be noted that TCS exposure during pregnancy may disrupt estrogen metabolism. The increased use of disinfectants during the 2020–2022 pandemic may have led to increased TCS exposure during pregnancy. Prenatal TCS exposure can interfere with key processes of neurodevelopment. It can therefore be speculated that the potential risk of ADHD in children born during this period may increase in the future, highlighting the need for long-term monitoring and epidemiological follow-up of birth cohorts. In addition, several studies published between 2023 and 2025 further showed that TCS concentrations and detection frequencies in water, indoor dust, human urine, and breast milk remained high after the pandemic, with exposure risks significantly higher than those reported before the pandemic.

In summary, TCS concentrations in various environmental media and biological samples showed an overall upward trend after 2019, which was likely related to the surge in the use of TCS-containing disinfectants and hygiene products during the COVID-19 pandemic (Chu et al., 2020; Kumar et al., 2021). This temporal pattern further highlights the urgency of research on the environmental risks and neurodevelopmental toxicity of TCS, and also provides a scientific basis for future strategies for the prevention and control of TCS pollution.

3.6. TCS and the nervous system

The nervous system plays an important role in regulating physiological and functional activities. By adjusting the functional activities of the body, it can adapt to the changing external environment and maintain a balance between the body and the external environment.

3.6.1. Interference with thyroid hormone levels

In a prospective birth cohort study, a significant negative correlation was observed between maternal urine TCS and cord blood-free triiodothyronine and maternal blood-free thyroxine concentrations during the third trimester of pregnancy (Wang, Ouyang, et al., 2017). An increase in urinary TCS concentrations (median TCS levels of 10.4 μg/g creatinine in adolescents and 11.5 μg/g creatinine in women) was associated with increased concentrations of total triiodothyronine in adolescents (Koeppe et al., 2013). In rodents, TCS exposure reduces thyroxine levels in pregnant females and offspring during pregnancy (Johnson et al., 2016; Paul et al., 2012), and the thyroid hormones transferred from the mother to embryo and fetus during pregnancy are critical for the normal development of their nerves (Haddow et al., 1999); even minor deficiencies in thyroid hormone concentrations during pregnancy have been associated with lower IQ scores in children (Escobar et al., 2004; Haddow et al., 1999; Henrichs et al., 2013; Korevaar et al., 2016). A recent study found that environmentally relevant concentrations of TCS could cause intergenerational thyroid hormone disruption in zebrafish, posing a threat to the ecological environment (Ding et al., 2025). These studies highlight the adverse effects of TCS on human brain health and the need for the continuous monitoring of its use and effects.

3.6.2. Hypothalamus and hippocampus

The adverse effects of TCS may be related to disturbances in the metabolic processes in the hypothalamus (Huang et al., 2021). After direct or indirect exposure to low-dose TCS, the transcriptomic profiles of the fetal hypothalamus of sheep during the third trimester of pregnancy was found to change drastically, with hypothalamic genes showing enrichment in pathways related to the cell cycle, reproductive processes, and feeding behavior, mainly associated with food intake and metabolism. If these changes persist in postpartum life, they may have adverse consequences in adulthood (Rabaglino et al., 2016).

TCS has a highly destructive effect on the functions of hippocampal neurons in rodents, and Tabari et al. (Shaghayegh Alizadeh Tabari et al., 2019) found that the morphological characteristics of neurons in the hippocampal region are changed after high-dose TCS exposure (2000 and 4000 mg/kg). Moreover, nuclear shrinkage and necrosis were evident, with the severe loss and necrosis of Purkinje cells and decreases in the total cortical thickness; however, Purkinje neurons are the only integrative centers of the cerebellar cortex and play a key role in coordination, neuromuscular strength, learning functions, and motor movement (Zhang et al., 2013).TCS alters the content and phosphorylation of synaptic plasticity-related proteins. Arias-Cavieres et al. (Arias-Cavieres et al., 2018) added low-concentration TCS (1–5 μM) to male rat hippocampal slices, impairing hippocampus-dependent spatial memory functions. Similar findings have been reported in the offspring of parent rats treated with TCS; axon degeneration was found to be strongly associated with altered hippocampal protein expression (Hao et al., 2022). TCS exposure was found to cause behavioral disorders in adolescent mice and mediate cellular inflammation by regulating STAT3 phosphorylation via PKM2 dimer (Wei et al., 2024).

3.6.3. Morphological changes in the nervous system

TCS may also affect neurobehavioral parameters in juvenile zebrafish (Wang, Wang, et al., 2019). When zebrafish were exposed to TCS at 0.6 mg/L (administered in the drinking water), the boundary between the midbrain and hindbrain of the zebrafish embryos disappeared, synaptic density decreased, axon length was reduced, and neuronal

continuity was disrupted. Apoptotic neurons and nerve pile vacuoles were observed in the brain, spinal cord, and caudal spinal cord. Embryos exposed to TCS at ≥ 0.4 mg/L show reduced body length and head size, curvature of the spine, and abnormal central nervous system (CNS) development (Kim et al., 2018). Muth-Köhne et al. found that exposure to the highest concentrations (2.8 μM) of TCS affects embryonic motor neuron development, resulting in edema; decreased blood circulation; and head, heart, and tail malformations (Muth-Köhne et al., 2012). When Pullaguri et al. (Pullaguri et al., 2020) treated zebrafish juveniles with 0.3 and 0.6 mg/L of TCS, an increase in the arcuate angle and discontinuity of neuromuscular connections, aberrations in skeletal muscle motor neuron innervation, and decreased tactile-induced escape responses in zebrafish larvae were observed. Similar findings in experiments by Liu et al. suggested that TCS can lead to enlarged ventricles in adult zebrafish, a decrease in the number of neuronal cells, formation of glial scarring, accompanied by abnormal glial cell proliferation, and abnormal hearing and vision in zebrafish (Liu et al., 2019). Using zebrafish larvae, Wang et al. found that developmental impairments caused by TCS included neuronal reduction, vascular ablation, and neuronal apoptosis (Wang et al., 2023). In addition, TCS may increase the risk of ischemic stroke through neurotoxic mechanisms in the central nervous system, including neuronal apoptosis and neuroinflammation (Cheng et al., 2025). All these studies confirm that TCS can cause morphological changes in the nervous system in animal models.

3.6.4. Oxidative stress

TCS induces oxidative stress, mainly by increasing the production of reactive oxygen species (ROS) (Lee et al., 2023). In a study of mouse neocortical neurons, increased ROS production was found to be dependent on N-methyl-D-aspartate (NMDA) (Szychowski et al., 2016). TCS can induce neurotoxicity by inducing DNA fragmentation and apoptosis in primary cultures of neocortical neurons through caspase-3, caspase-8, and Fas receptor signaling (Park et al., 2016; Ruszkiewicz et al., 2017; Szychowski et al., 2015, 2019; Wang, Liu, et al., 2021). Park et al. exposed rat neural stem cells to TCS and found that compared with the control group, 50 μM TCS did not affect expression of the MAPK signaling proteins themselves but induced increased expression of phosphorylated p38 and JNK proteins to varying degrees, while reducing the expression of phosphorylated ERK, Akt, and PI3K proteins; moreover, dysregulation of the phosphorylation of MAPK signaling proteins may contribute to premature apoptosis (Park et al., 2016). Li et al. reported similar findings when studying the neurotoxic effect of TCS on rat pheochromocytoma PC12 cells. Here, the production of ROS was increased in TCS-treated PC12 cells, at concentrations of 1 and 10 μM; moreover, the inhibition of Akt and mTOR phosphorylation was observed in PC12 cells treated with 10 and 50 μM TCS, and the p38 MAPK pathway was activated in the 50 μM TCS treatment group. However, TCS did not affect AMPK phosphorylation in these cells (Li et al., 2019). Taken together, these results suggest that TCS causes adverse reactions and oxidative stress in PC12 cells, accompanied by Akt/mTOR inhibition and p38 activation. In addition, when mouse cortical neurons were exposed to TCS for 48 h, peroxisome proliferator-activated receptor gamma expression was increased and nuclear factor kappa-light-chain-enhancer of activated B cells expression was decreased, accompanied by high levels of ROS, resulting in increased expression of nitric oxide synthase (NOS) and decreased expression of neuronal NOS (nNOS) (Szychowski et al., 2021). Further, Bao et al. showed that TCS interferes with the Nrf2/HO-1-mediated oxidative stress response (Bao et al., (2020) and induces oxidative damage in HT-22 neurons, and activation of the PI3K/Akt/JNK/Nrf2/HO-1 signaling cascade was determined to be the main mechanism of TCS-induced HT-22 neuronal toxicity. Importantly, aged PE-MPs containing triclosan will exacerbate oxidative stress and neurotoxicity in Xenopus tropicalis tadpoles (Lin et al., 2024). Wang et al. and others (Wang, Huang, et al., 2019; Wang et al., 2020; Yoon et al., 2017) also identified TCS-induced activation of the Nrf2 signaling pathway in zebrafish brain tissue and human mesenchymal stem cells. TCS has been shown to interact with aryl hydrocarbon receptors (AhRs) in mouse neurons (Szychowski et al., 2016). Szychowski et al. (Szychowski et al., 2020) elucidated the mechanism by which TCS stimulates ROS production, measuring ROS levels in neuronal cells co-treated with α naphthoflavone (αNF) or β naphthoflavone (βNF). Both agonists (βNF) and antagonists (αNF) of AhR can significantly reduce TCS-induced ROS production, increases in LDH release, and caspase-3 activity. Both αNF and βNF inhibit TCS-stimulated xanthine oxidase (XO) activity. Therefore, XO may be responsible for the ROS production induced by TCS. However, the mechanisms associated with βNF, αNF, and TCS require further study, and recent experiments have found that TCS increases the expression of Sirt1, Sirt3, and AhR proteins in murine neurons cultured in the cerebral cortex; moreover, the increase in the expression of these proteins may be the result of an increased response to ROS production or the response to AhR stimuli (Szychowski et al., 2022). Acetylcholinesterase (AChE) is a cholinergic enzyme, and any alteration in its activity is a recognized neurotoxic biomarker. Using adult zebrafish as subjects, TCS was found to induce neurotoxicity at sublethal concentrations by inducing oxidative stress, thereby inhibiting AChE *in vivo* (Pullaguri et al., 2019, 2020), at lower concentrations (0.6 mg/L) (Pullaguri et al., 2022), contradicting previous studies showing that exposure to TCS at very low concentrations (50–100 μg/L) can lead to increased AChE activity in zebrafish and other species (Falisse et al., 2017; Gomes et al., 2021). The differences among these studies may reflect a concentration-dependent biphasic response. Studies have shown that at relatively high concentrations, TCS directly inhibits AChE activity, and this effect is further enhanced by oxidative stress. Conversely, the increase in AChE activity at lower TCS concentrations may represent an early compensatory response of the cholinergic system rather than a completely opposite mechanism. This nonlinear regulation may also interact with CNR1-mediated dopamine signaling. Since CNR1 modulates presynaptic neurotransmitter release and indirectly regulates dopamine output via GABA and glutamatergic circuits, the changes in cholinergic tone may alter local neural network activity, thereby influencing downstream dopaminergic effects associated with CNR1 signaling (Wilson, 2002). Therefore, trace amounts of TCS in tissues can be harmful. Using animal models, Sahu (Sahu et al., 2018), Sager (Sager et al., 2021), Huang (Huang et al., 2023), and others observed that AChE activity in animal brains is inhibited after TCS exposure. Therefore, it is speculated that, because AChE activity strictly regulates the amount of acetylcholine available at cholinergic synapses, an increase or decrease in its activity will lead to abnormal acetylcholine levels, resulting in abnormal neurotransmission. The main pathways of oxidative stress induced by triclosan are shown in Fig. 4. However, further research is needed to determine how TCS leads to increased AChE activity, particularly from the perspective of post-translational modifications.

3.6.5. Neurotoxicity and microRNA (miRNAs)

miRNAs are small non-coding RNAs containing 18–25 nucleotides that are involved in the post-transcriptional regulation of gene expression. miR-219, an evolutionarily conserved miRNA, is one of the miRNAs involved in TCS-induced toxicity, confirming the significance of this breakthrough in understanding the toxic effects of TCS on myelin (Zhao, He, et al., 2010). The disruption of myelin during development can lead to neuronal differentiation, proliferation, migration, gliogenesis, synapses, dendritic growth, myelination, apoptosis, and synaptic pruning, thereby increasing the risk of behavioral disorders or cognitive deficits (de de Graaf-Peters and Hadders-Algra, 2006; Rice and Barone, 2000). After TCS exposure, the abnormal expression of marker genes related to nerve cell migration, axonal orientation, and myelination in zebrafish significantly increases expression of the mitochondrial apoptosis activator Bcl-2l11 and the functional knockdown and overexpression of CELF2, confirming that TCS leads to neurological damage and developmental defects through CELF2-mediated gene-selective splicing

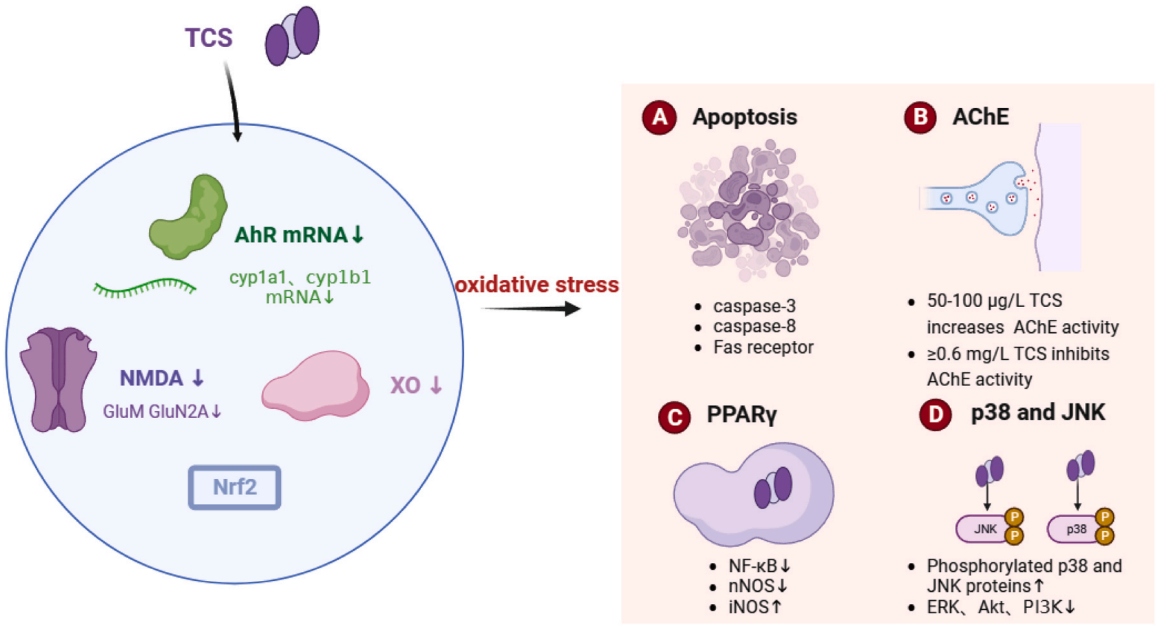


Fig. 4. The main pathways of oxidative stress induced by triclosan. TCS can downregulate the expression of aryl hydrocarbon receptor (AhR) mRNA, cytochrome P450 family genes (cyp1a1, cyp1b1) mRNA, N-methyl-D-aspartate receptors (NMDA, including GluM and GluN2A subunits), and xanthine oxidase (XO), activate the Nrf2 pathway and induce cellular oxidative stress. The right side illustrates four types of downstream effects mediated by oxidative stress: A. Apoptosis; B. Dual-directional regulation of acetylcholinesterase (AChE) activity; C. Modulation of the PPARγ pathway; D. Imbalance of the mitogen-activated protein kinase (MAPK) pathway.

events (Wang, Wang, et al., 2021). Moreover, TCS exposure increases apoptosis in the CNS cells of zebrafish, affecting the structure of the CNS and resulting in reduced synaptic density and axon length (Kim et al., 2018). Wang et al. constructed an AOP framework for TCS based on the data, and analyzed 29 key cirRNA events and miRNA binding in zebrafish exposed to TCS (Wang, Han, et al., 2024). Further research has revealed that triclosan can induce neurotoxicity through the circRNA266/miR-214/dapk2-gad2 axis (Wang et al., 2025). These miRNA studies provide new insights into the epigenetic regulatory mechanisms of TCS-induced toxicity, supporting the study of TCS adverse effects, and other mechanisms of influence involved are shown in Table S5.

3.6.6. *Neurodevelopmental disorders in offspring*

In addition to damaging the nervous system, TCS exposure affects neurodevelopment in offspring. Epidemiological investigations have shown that perinatal exposure to TCS causes neurodevelopmental impairment in fetuses or children of a certain age, as detailed in Table S6. Prenatal exposure to TCS is associated with reduced attention in infancy and slower information processing in male infants (Cragoe et al., 2024). In cohort studies, TCS was associated with significantly lower cognitive test scores in 8-year-olds (Jackson-Browne et al., 2018) and mental functions in 7-year-olds (Braun et al., 2017), but the adverse effects related to behavioral problems were more pronounced in boys (Lei et al., 2024), with a small significant association with visuospatial ability in 8-year-olds (Tanner et al., 2020). Tran et al. (Tran, Jung, et al., 2020) exposed female mice to TCS during pregnancy and found that apoptotic signals in the brains of adult offspring mice was increased, levels of aspartate-specific cysteine proteinase 3 were elevated, level of the anti-apoptotic Bcl-2 protein was reduced, and ratio of brain weight to total body weight was lower in the TCS group than in the control group. In behavioral experiments, mice in the TCS treatment group were found to have cognitive dysfunction, social deficits, and anxiety-like behaviors in the behavioral fields of spatial and non-spatial learning and memory. Neuronal retinoic acid (RA) signaling plays an important role in mediating neuronal differentiation, synaptic plasticity, and neuronal tissue formation. Manipulation of the RA signaling pathway has been shown to alter behavior and induce autism-like behaviors in offspring (Bremner and McCaffery, 2008). Hao et al. (Hao et al., 2019) found that the offspring of rats exposed to TCS (50 mg/kg/day) during pregnancy exhibited neuronal RA signaling disruptions in their brains, which could impair RA signaling or RA-mediated neuronal synaptic plasticity through events downstream of RA synthesis, thereby inducing autistic behaviors. In addition, toxicological threshold data from animal studies were summarized by exposure window and species, as shown in Table S7. In conclusion, in animal models, maternal exposure to TCS has been shown to cause neurodevelopmental disorders in offspring and affect their neurobehavior.

3.7. *AOP construction*

Based on the literature review of TCS-induced neurodevelopmental toxicity, TCS exposure was found to be associated with ADHD-like behaviors in offspring. To investigate its potential mechanisms, we constructed a putative AOP framework for TCS-induced ADHD. A schematic of the workflow of this study is presented in Fig. S1. We searched CTD, GeneCards, DisGeNET and MalaCards databases and obtained a large amount of genes and phenotypes data, which were subjected to bioinformatics analyses (KEGG and GO) in order to further obtain the interactions among stressors, target genes, target phenotypes and disease. Finally, build a network of stressors-gene-phenotype-disease relationships to elucidate the intrinsic biological mechanism of TCS-induced ADHD.

3.7.1. *Gene and phenotype search results*

The extracted related genes and phenotypes data are shown in Table S8. We extracted 3470 TCS-related genes and 3041 phenotypes in CTD and GeneCards database. In addition, we also retrieved 953 ADHD-related genes and 3325 phenotypes in DisGeNET and MalaCards databases, respectively.

3.7.2. *Identification of target genes*

After excluding overlapping genes in each gene set, we collected a total of 4185 genes associated with TCS or ADHD. According to the gene

screening method described in 2.2.1, we identified 46 genes as target genes for this study (Fig. S2). Ultimately, information was reviewed to verify whether the 46 target genes were associated both TCS and ADHD, and to confirm the set of target genes. We defined these genes as potential MIEs in the AOP network of TCS-induced ADHD.

3.7.3. Identification of target phenotypes

We collected a total of 5373 phenotypes associated with TCS or ADHD. We performed GO and KEGG enrichment analyses on the target genes identified in 3.7.2 and obtained 594 biological process (BP), 32 cellular components (CC), 54 molecular function (MF) and 19 KEGG pathways associated with 46 genes. The 699 phenotypes derived from the enrichment analysis were subjected to Venn analysis with the acquired TCS-related and ADHD-related phenotypes. The 83 phenotypes present simultaneously in the three sets were identified as target phenotypes (Fig. S3). Also, according to the hierarchical relationship of GO terminology, the target phenotypes were categorized into three levels: subcellular, cellular and system.

3.7.4. TCS-gene-phenotype-ADHD network

The TCS-Gene-Phenotype-ADHD network (Fig. 5) was mapped using the obtained target genes and target phenotypes data. There are 122 nodes in this network relationship graph, including 1 stressor (TCS), 1 AO (ADHD), 37 target genes, 79 GO Terms and 4 KEGG pathways. The interrelationships between these 122 nodes generated 712 edges. The number of edges of a node in a Cytoscape network is measured by "Degree", and nodes with more degrees have more evidence to play a central role in the network. In this network, the degrees of "TCS" and "ADHD" are both 120, which are the largest among all nodes. Therefore, we can determine that all target genes and target phenotypes are associated with triclosan and ADHD. Among the gene nodes, CNR1 (Degree: 36) and AKT1 (Degree: 35) had the highest degree. "Modulation of chemical synaptic transmission" (P21) had the highest number of edges (Degree: 19) among all phenotype nodes.

3.7.5. cAOP network of TCS- induced ADHD

Based on the TCS-Gene-Phenotype-ADHD network, we filtered the phenotypes highly correlated with ADHD based on the degree value ranking. The target phenotypes of GO terms were categorized into three event levels (subcellular, cellular, and system) based on their hierarchical relationships. KEs at the subcellular level included "modulation of chemical synaptic transmission" (GO:0050804, P21), "regulation of neurotransmitter levels" (GO:0001505, P2), etc. Relevant phenotypes at the cellular level are "synaptic membrane" (GO:0097060, P52), "neuron death" (GO:0070997, P51) and "cellular calcium ion homeostasis" (GO:0006874, P29). At the system level, we screened for phenotypes related to "cognition" (GO:0050890, P76), "learning or memory" (GO:0007611, P60), and "response to nutrient levels" (GO:0031667, P70). In addition, two KEGG pathways were identified as "Neuroactive ligand-receptor interaction" (KEGG:hsa04080, P81), and "cAMP signaling pathway" (KEGG:hsa04024, P80). 35 genes associated with these 12 phenotypes were identified as MIEs. Based on the above gene-phenotype interactions and the logic of the biological hierarchy, we constructed a cAOP network for TCS-induced ADHD (Fig. 6A).

3.7.6. CNR1 induced AOP of ADHD

Based on the cAOP network we constructed, CNR1 was the most enriched among the phenotypes at each level compared with other MIEs (Table S10), therefore CNR1 was used as the MIE. According to the enrichment results, we choose "Neuroactive ligand-receptor interaction" as the key pathway for constructing AOP (Fig. 6B). When the human body is exposed to TCS, the CB1 receptor encoded by CNR1 is activated, which affects intracellular Ca^{2+} homeostasis through the pathway of "Neuroactive ligand-receptor interaction", thus regulates chemical synaptic transmission and neurotransmitter levels, and ultimately causing cognitive, learning and memory dysfunction in the organism. To further validate the key event relationship (KER), we searched the relevant literature in the Web of Science and PubMed databases to assess the weight of evidence (WoE) for the KER according to the Organization for Economic Cooperation and Development Handbook of AOPs (Users, 2025) (Table S13, S14). Therefore, based on empirical and epidemiologic evidence, the necessity of the AOP involving KE was rated as "high". Meanwhile, the biological plausibility as well as the empirical support for KER was rated "High". In conclusion, the AOP for

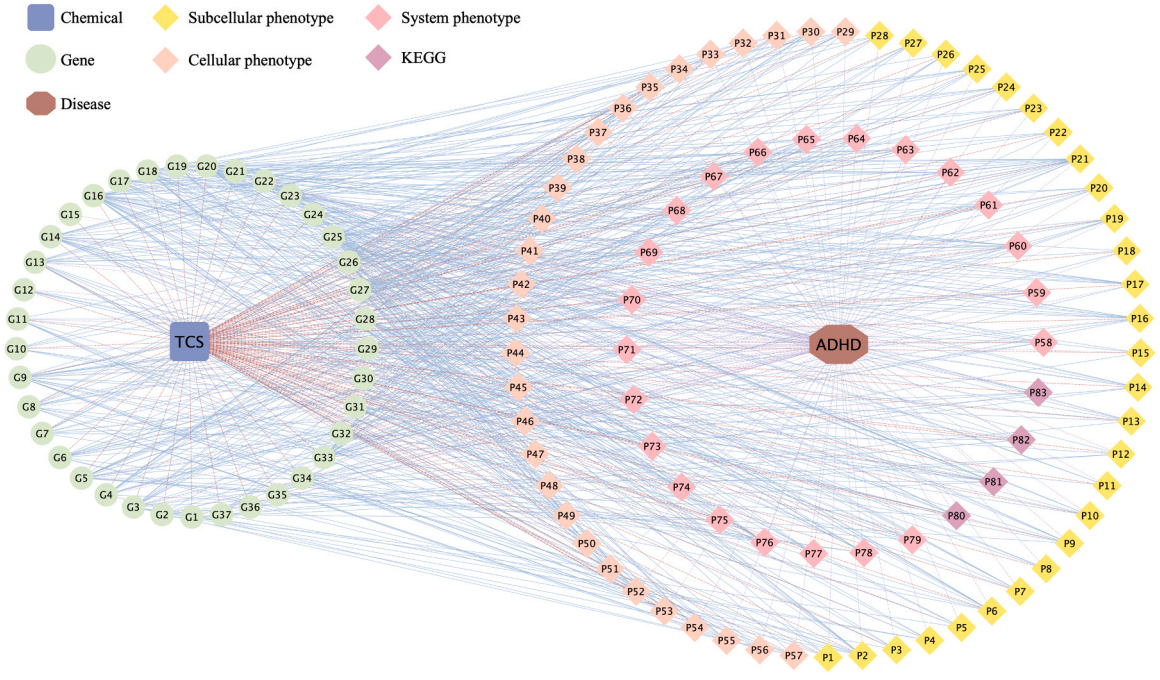


Fig. 5. TCS-Gene-Phenotype-ADHD network. Rectangle blue node indicates TCS; Round green nodes indicate target genes (G1-G37); Rhombic node indicate phenotypes (subcellular phenotypes: P1-P28; cellular phenotypes: P29-P57; system phenotypes: P58-P79; KEGG pathways: P80-P83); Octagonal brown node indicates ADHD. More detailed information on genes and phenotypes is shown in Table S9-S11.

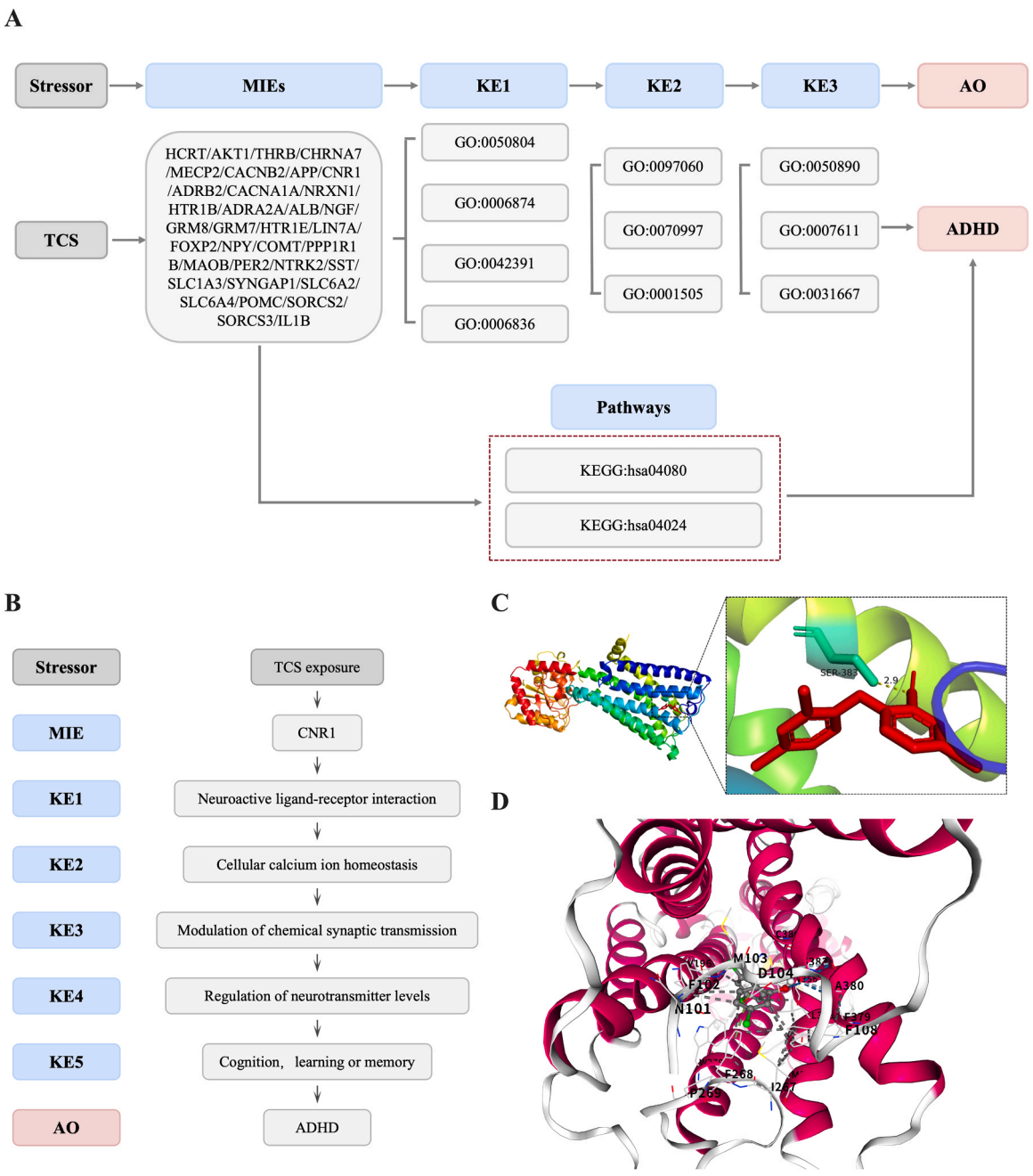


Fig. 6. **A**: cAOPs network from TCS to ADHD, including 10 GO terms and 2 KEGG pathways were depicted as intermediate KEs. 10 GO terms were organized into three levels (subcellular, cellular, system) and connected by their relationships. Genes related to these 12 KEs were selected as the initial targets. Thus, a cAOPs network connecting TCS and ADHD was generated. **B**: CNR1-initiated AOP of ADHD. **C**: The outcomes of molecular docking between TCS and CNR1, binding energy −7.8 kcal/mol. **D**: 2D interaction diagram of TCS-CNR1 binding complex. TCS forms hydrogen bond with CNR1 residue Ser383 and hydrophobic contacts with Phe174, Trp279 and Leu359 (PDB ID: 5U09, resolution: 2.80 Å).

CNR1-induced ADHD has high plausibility. Based on the previous review of the toxicity mechanisms of TCS, we developed a conceptual diagram to illustrate its connection with the core CNR1-AOP (Fig. 7).

3.8. Molecular docking

The key MIE of the AOP framework is validated using molecular docking. TCS and CNR1 have a strong binding force. The results are shown in the Fig. 6C-D.

4. Discussion

In this study, we provide a comprehensive overview of the current status of environmental contamination with TCS and focus on the potential neurological threats associated with exposure to this compound. Using TCS-induced ADHD as an example, we used an adverse outcome pathway (AOP) framework to construct a possible mechanism network. TCS exposure in humans affects the "neuroactive ligand–receptor interaction" pathway through CNR1, regulates calcium homeostasis, and modulates chemical synaptic transmission and neurotransmitter (NT) levels, leading to cognitive, learning, and memory deficits and ultimately ADHD-like behaviors.

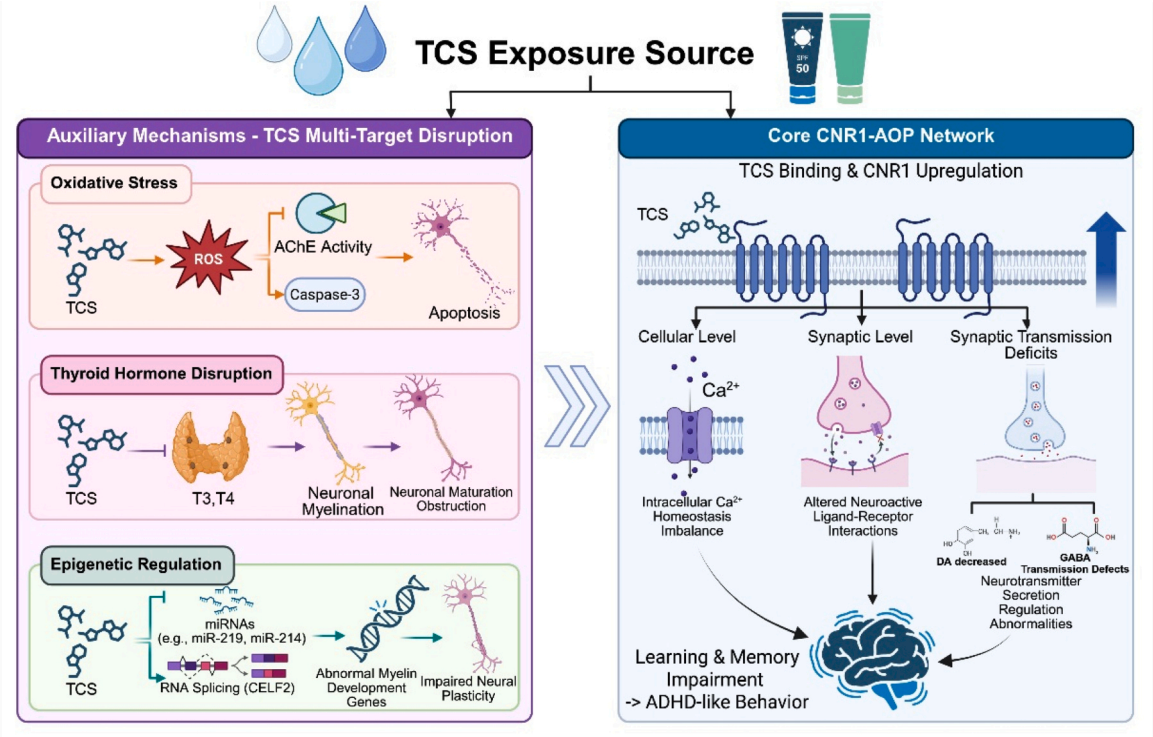


Fig. 7. Conceptual diagram linking auxiliary mechanisms to the core CNR1-mediated adverse outcome pathway (AOP) for TCS-induced ADHD. TCS exposure activates three auxiliary pathways: oxidative stress (exacerbates neuronal oxidative injury), thyroid hormone disruption (impairs neuronal maturation and myelination), and epigenetic regulation (alters neural plasticity). These pathways synergize with the core CNR1-AOP: TCS binds to CNR1 (binding energy −7.8 kcal/mol), activates the "neuroactive ligand-receptor interaction" pathway, disrupts chemical synaptic transmission and intracellular calcium homeostasis, and ultimately induces ADHD-like behaviors (cognitive deficits, inattention, hyperactivity). This diagram clarifies how auxiliary mechanisms amplify CNR1-mediated neurotoxicity.

TCS is a chemical bacteriostatic agent that is widely used in daily necessities, such as antibacterial hand sanitizers, toothpastes, and children's toys. TCS and its metabolites can be absorbed directly or indirectly through the mucous membranes of the skin, respiratory tract, and digestive tract and can be present in daily necessities or accumulate in water, soil, and organisms. Currently, TCS has been detected in all organisms worldwide. Based on the further study of TCS, compared with similar products without TCS, not only do TCS-containing products lack obvious bacteriostatic effects, but also, long-life TCS exposure may cause a variety of damaging effects on the human body (Chrz et al., 2023). In the last decade, studies have found that people in 25 countries, including China, face simultaneous exposure to EDCs such as PAE, BPA, TCS, and PFAS, and an association between these compounds and miscarriage is possible (Xu et al., 2024). TCS, an endocrine-disrupting chemical, poses a silent threat to human health. After COVID-19, the overuse of disinfection products containing antimicrobials such as TCS has led to a significant increase in TCS concentrations in the environment and human exposure levels (Luo et al., 2024). Concerns exist that it may interfere with hormonal functions (Đurić et al., 2024). In addition, we found that it may have a synergistic toxicity mechanism with nanoplastics and MPs and other endocrine disruptors (Zong et al., 2024), which has caused further consideration of its effects.

Co-exposure to multiple pollutants is very common in the environment. The combined effects of TCS with MPs, BPA, and other contaminants may amplify the risk of neurodevelopmental toxicity. Previous studies have shown that co-exposure to TCS and microplastics, such as PP and PS, can alter acetylcholinesterase activity and 5-HT levels in zebrafish brain tissue by promoting TCS bioaccumulation and enhancing oxidative stress. In addition, such co-exposure may induce neuroinflammation through the gut-brain axis, thereby exacerbating cognitive dysfunction and synaptic transmission deficits (An and Zhang, 2023; Liu et al., 2025). Co-exposure to TCS and BPA may also disrupt neuronal maturation and myelination by synergistically interfering with thyroid hormone signaling and competitively binding to neuronal receptors. This interference could impact the development of the thyroid–pituitary–hypothalamus axis, potentially leading to pregnancy loss (Xu et al., 2024; Zong et al., 2024). These synergistic effects suggest that the toxicity assessment of a single TCS may underestimate the harm of actual environmental exposure. In the future, it is necessary to explore its combined effects on neural development in combination with multi-pollutant coexistence scenarios.

In recent years, AOP has emerged as a new tool in computational toxicology, as it is faster and more accurate and uses fewer laboratory resources for assessing the risks of compounds human health. The framework utilizes computational toxicology to construct a network of MIE or KE relationships between compounds and diseases based on available relevant toxicity data (Zhang et al., 2022), and it can be used to assess the reliability of AOP based on the weight of evidence. Therefore, we collected relevant data from CTD, GeneCards, DisGenNET, and MalaCards to construct a cAOP network and obtained a preliminary AOP framework for ADHD with high confidence.

The AOP framework further explains the mechanism by which TCS leads to cognitive, learning, and memory deficits, ultimately triggering ADHD and facilitating toxicity risk assessment associated with TCS-induced ADHD. *CNR1* encodes the CB1 receptor, the most abundant G protein-coupled receptor in the mammalian brain. The CB1 receptor mediates a series of signaling events in the brain, including the closing of Ca^{2+} channels, opening of K^{+} channels, inhibition of adenylyl cyclase activity, and stimulation of protein kinases. These signals regulate synaptic transmission and neuronal gene expression (Piomelli, 2003). Molecular docking results showed strong binding between TCS and CNR1, and the CTD further indicates that TCS increases CNR1 mRNA expression. Thus, we suggest that CNR1 is a potential molecular target for TCS-induced ADHD. The "neuroactive ligand–receptor signaling

pathway" pathway plays an important role in the mechanism underlying TCS-induced ADHD (Hu et al., 2025). LPS exposure affected the glutamatergic synapse and neuroactive ligand-receptor interaction, which were associated with early life inflammation-induced cognitive impairment (Wu et al., 2023). These results suggest that TCS exposure may activate CNR1, which modulates synaptic chemical transmission and regulates NT levels through the "neuroactive ligand–receptor signaling pathway". In addition, abnormal CNR1 expression affects intra- and extracellular Ca^{2+} concentrations (Andrade et al., 2019), shutting down Ca^{2+} channels and inhibiting glutamate release. The CB1 receptor is predominantly present in presynaptic neurons, and the activated CB1 receptor reduces presynaptic Ca^{2+} influx through voltage-gated calcium channels(Boczek and Zylinska, 2021). The calcium ion signaling pathway is associated with the risk of ADHD, ASD, and other neurodevelopmental disorders (Nagai et al., 2019). Previous studies suggested that synaptic transmission is calcium ion-dependent. Changes in chemical synaptic transmission alter transmitter release and effective neurotransmitter availability at synapses (Sudhof, 2011). Importantly, neurons conduct electrochemical signals via NTs, accounting for most of the cellular signaling in the brain (Hyman, 2005). The CB1 receptor, encoded by *CNR1* and predominantly expressed in presynaptic neurons, inhibits the release of various NTs, including GABA, glutamate, and dopamine, into the synapse, ultimately leading to cognitive and memory deficits (Wilson, 2002). A previous study confirmed that chronic TCS exposure downregulates DRD2 protein expression, inhibits dopamine release in the PFC, and induces ADHD-like behaviors in mice (Cui et al., 2024). Based on this AOP, it can be assumed that TCS exposure promotes CNR1 protein expression, which in turn affects dopamine release. Yao et al. found significant associations between the *CNR1* gene and psychiatric disorders, such as ADHD, and that CB1 receptor dysfunction alters several behavioral responses and increases susceptibility to anxiety and mood disorders, such as depression and aggression (Yao et al., 2018). These findings support our proposed mechanism, although the specific mechanisms of action and key event relationships require further exploration.

It should be noted that molecular docking was used solely to preliminarily assess the feasibility of the AOP framework. The present study does not provide direct experimental evidence that TCS binds to or activates CNR1 as a validated molecular initiating event. Further studies are required to validate the docking results, and CNR1 will be considered a potential target in our future work. Meanwhile, the development and utilization of the AOP have been improved and promoted, providing a basis for researchers to conduct subsequent related work.

5. Conclusion

In summary, this study provides a systematic overview of environmental and human TCS exposure in recent years and explores the related exposure risks before and after COVID-19. Therefore, the presence of TCS and other related micropollutants in the environment deserves further attention. AOP, an emerging toxicity risk assessment tool in recent years, has promising applications in the field of alternative toxicology. For the first time, we developed a CNR1-mediated AOP framework for TCS–ADHD, which provides new potential targets for exploring the mechanism of TCS-induced neurotoxicity. In conclusion, our study provides new insights into human health risk assessments associated with TCS exposure and interventions for the treatment of related diseases.

6. Future perspectives

Human exposure to TCS has become a widespread public health concern. However, the adverse health effects of TCS exposure are unclear. In future, attention should be paid to the following aspects:

1) **Increased environmental exposure to TCS after the epidemic warrants greater attention to its toxic effects.** In recent years, various studies have shown that TCS concentrations in the environment have surged since the major COVID-19 outbreak, posing various threats to other biological populations. Although no reports have indicated markedly elevated concentrations to which humans are exposed following the outbreak, the potential toxic effects on human health cannot be ignored.
2) **Development of relevant products and technologies to regulate environmental contaminants.** The real-time monitoring of TCS concentrations in the environment is an important prerequisite for studying its toxicity mechanism. An urgent need exists to develop products or tools suitable for monitoring TCS concentrations across different environmental media. Therefore, research on TCS removal technologies in the environment should not be delayed.
3) **The mechanisms underlying toxic effects are complex, necessitating the application of a new generation of computational toxicology tools for in-depth investigation.** The mechanisms underlying TCS toxicity in humans and other organisms remain unclear. In future, it will be feasible to conduct more in-depth studies using emerging alternative toxicological methods instead of animal testing.
4) **Co-exposure between compounds may result in synergistic toxicity.** Previous studies have found that co-exposure to TCS and other endocrine-disrupting compounds is widespread in organisms. The occurrence of more threatening synergistic toxic effects therefore needs to be further explained, and the potential adverse health effects require urgent attention.

CRediT authorship contribution statement

Ziyun Wei: Visualization. **Fu Liu:** Resources. **Yuxuan Peng:** Data curation. **Chang Shu:** Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Linlin Zheng:** Investigation. **Jianing Shang:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **He Cui:** Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Xiao Ni:** Resources. **Jieyu Liu:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Conceptualization.

Consent for publication

The authors declare that this manuscript does not contain any individual person's data and material in any form.

Consent to participate

All participants agreed to participate in this study and signed the informed consent.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2026.120169.

Data availability

No data was used for the research described in the article.

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Glossary

AChE: acetylcholinesterase
ADHD: attention-deficit/hyperactivity disorder
AhRs: aryl hydrocarbon receptors
AO: adverse outcome
AOP: adverse outcome pathway
ASD: autism Spectrum Disorder
BP: biological process
CC: cellular component
CNR1: cannabinoid receptor 1
CNS: central nervous system
COVID-19: coronavirus disease 2019
FDA: Food and Drug Administration
GO: Gene Ontology
KE: key event
KEGG: Kyoto Encyclopedia of Genes and Genomes
KER: key event relationship
MF: molecular function

MIE: molecular initiating event
miRNAs: microRNAs
MPs: microplastics
nNOS: neuronal nitric oxide synthase
NOS: nitric oxide synthase
NT: neurotransmitter
PPCPs: pharmaceuticals and personal care products
RA: retinoic acid
RGO: reduced graphene oxide
ROS: reactive oxygen species
RW: reclaimed water
TCS: triclosan
USEPA: United States Environmental Protection Agency
WWTPs: wastewater treatment plants
XO: xanthine oxidase
αNF: α-naphthoflavone
βNF: β-naphthoflavone